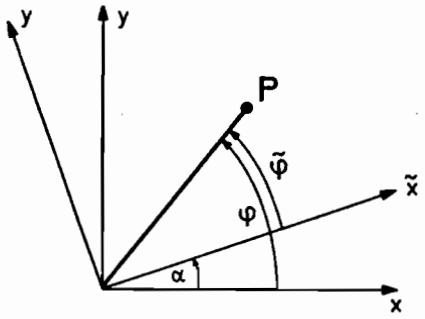
APPENDIX A

Angular Momentum Algebra in the Laboratory and the Body-Fixed System

Throughout the book we use the conventions and phases of Edmonds for angular momentum coupling [Ed 57]. The only place where we have to enter a little into details is the case of the transformation of angular momenta into the body-fixed system. We shall carry out this transformation in the sense of Edmonds [Ed 57, Chap. IV].

To study the transformation properties of wave functions in Hilbert space under a rotation of the coordinate system, we shall first investigate a simple one-dimensional example.



Let $f(\varphi)$ be a function, which only depends on the azimuth angle with respect to the z-axis of the system. For a rotation of this system through the angle α the azimuth of a fixed point P in space transforms into $\tilde{\varphi} = \varphi - \alpha$. Since the value of the function f at the fixed point P is not changed by a transformation of the coordinate system, the function f in the

old system transforms into a new function $\tilde{f}$ in the new system with the property

$$\tilde{f}(\tilde{\varphi}) = f(\varphi) = f(\tilde{\varphi} + \alpha) = e^{i\alpha J_z} f(\tilde{\varphi}) \tag{A.1}$$

or, if we change the nomenclature and use φ instead of $\tilde{\varphi}$,

$$\tilde{f}(\varphi) = e^{i\alpha J_z} f(\varphi) \qquad \text{with } J_z = \frac{1}{i}\frac{\partial}{\partial \varphi}. \tag{A.2}$$

In the general case of a three-dimensional rotation we use the *Euler angles* $\Omega = (\alpha, \beta, \gamma)$ as defined in [Ed 57, Chap. I]: Under a rotation of the coordinate frame around the angles Ω the wave function $|\Psi\rangle$ is transformed to

$$|\tilde{\Psi}\rangle = R(\Omega)|\Psi\rangle \tag{A.3}$$

with the unitary *rotational operator*

$$R(\Omega) = e^{i\alpha J_z} e^{i\beta J_y} e^{i\gamma J_z}, \tag{A.4}$$

where J_x, J_y, J_z act on the coordinates $\mathbf{x}_1, \ldots, \mathbf{x}_A$. As discussed in [Ed 57], we can also include spin coordinates. $\mathbf{J}$ is then the sum of orbital angular momentum and spin.

Corresponding to the transformation (A.3) of vectors in the Hilbert space, we get a *transformation* for operators T:

$$T \to \tilde{T} = RTR^+. \tag{A.5}$$

By definition, for a *spherical tensor operator* $T_{\lambda\mu}$ [Ed 57, Eq. (5.2.1)] we find

$$\tilde{T}_{\lambda\mu} = R(\Omega) T_{\lambda\mu} R^+(\Omega) = \sum_{\mu'} D^{\lambda}_{\mu'\mu}(\Omega) T_{\lambda\mu'}, \tag{A.6}$$

where the *Wigner functions* $D^{\lambda}_{\mu'\mu}(\Omega)$ are given by

$$D^{\lambda}_{\mu'\mu}(\Omega) = \langle \lambda\mu' | R(\Omega) | \lambda\mu \rangle. \tag{A.7}$$

We now wish to investigate a rotation Ω from the laboratory system (axis x, y, z) to a body-fixed system (axis $1, 2, 3$) at some fixed time t. Keeping in mind that the angular momentum operators J_x, J_y, J_z are generators for the infinitesimal rotations around the axis x, y, z, we can represent these operators as differential operators in the Euler angles α, β, γ. For instance, we can achieve a rotation around the z-axis by an amount $\delta\varphi$ by diminishing the Euler angle α by $\delta\varphi$ without changing the body-fixed axis, that is, *

$$L_z = -\frac{1}{i}\frac{\partial}{\partial \alpha} \triangleq J_z. \tag{A.8}$$

*In the following we shall denote the angular momentum operators expressed in terms of Euler angles by L_k ($k = x, y, z$). However, this does not mean that these operators only represent orbital angular momenta. The minus sign in Eq. (A.8) deviates from the usual representation of angular momenta in Euler angles [Ed 57, Eq. (2.2.2)]. The same change of sign occurs in the definition of the collective momentum operator $P = -(\hbar/i)(\partial/\partial q)$ in Eq. (10.92).

In the same way, from the definition of the Euler angles [Ed 57] we get:

$$-\frac{1}{i}\frac{\partial}{\partial\beta} \triangleq J_{y'} = e^{i\alpha J_z} J_y e^{-i\alpha J_z} = \cos\alpha\ J_y + \sin\alpha\ J_x \tag{A.9}$$

and

$$-\frac{1}{i}\frac{\partial}{\partial\gamma} \triangleq J_{z''} = e^{i\alpha J_z} e^{i\beta J_y} J_z e^{-i\beta J_y} e^{-i\alpha J_z}$$

$$= \cos\beta\ J_z - \sin\beta(\cos\alpha\ J_x - \sin\alpha\ J_y). \tag{A.10}$$

The inversion of (A.8)–(A.10) gives

$$\begin{aligned} L_x &:= \frac{1}{i}\left(-\cos\alpha\ \mathrm{ctg}\ \beta\frac{\partial}{\partial\alpha} - \sin\alpha\frac{\partial}{\partial\beta} + \frac{\cos\alpha}{\sin\beta}\frac{\partial}{\partial\gamma}\right) \triangleq J_x, \\ L_y &:= \frac{1}{i}\left(\sin\alpha\ \mathrm{ctg}\ \beta\frac{\partial}{\partial\alpha} - \cos\alpha\frac{\partial}{\partial\beta} - \frac{\sin\alpha}{\sin\beta}\frac{\partial}{\partial\gamma}\right) \triangleq J_y, \\ L_z &:= \frac{1}{i}\left(-\frac{\partial}{\partial\alpha}\right) \triangleq J_z. \end{aligned} \tag{A.11}$$

These are components of the angular momentum **J** with respect to the axis x, y, z in the laboratory system. In the same way we can calculate the components of **J** with respect to the body-fixed axis 1, 2, 3. To do this we use the fact that the spherical components L_μ, defined by

$$(L_\mu) = (L_{-1}, L_0, L_{+1}) := \left(\frac{1}{\sqrt{2}}(L_x - iL_y), L_z, \frac{-1}{\sqrt{2}}(L_x + iL_y)\right), \tag{A.12}$$

form a spherical tensor of rank one. The spherical components of the angular momentum with respect to the body-fixed axis are therefore

$$I_\mu = \sum_{\mu'} D^1_{\mu'\mu}(\Omega) L_{\mu'} = \sum_{\mu'} L_{\mu'} D^1_{\mu'\mu}(\Omega). \tag{A.13}$$

With the explicit form of $D^1_{\mu'\mu}(\Omega)$ [Ed 57, Eq. 4.1.15], after transforming back to cartesian components we gain

$$\begin{aligned} I_1 &= \frac{1}{i}\left(-\frac{\cos\gamma}{\sin\beta}\frac{\partial}{\partial\alpha} + \sin\gamma\frac{\partial}{\partial\beta} + \cos\gamma\ \mathrm{ctg}\beta\frac{\partial}{\partial\gamma}\right), \\ I_2 &= \frac{1}{i}\left(-\frac{\sin\gamma}{\sin\beta}\frac{\partial}{\partial\alpha} - \cos\gamma\frac{\partial}{\partial\beta} + \sin\gamma\ \mathrm{ctg}\beta\frac{\partial}{\partial\gamma}\right), \\ I_3 &= \frac{1}{i}\left(-\frac{\partial}{\partial\gamma}\right). \end{aligned} \tag{A.14}$$

The components of **J** in the body-fixed system have the following properties.

(i) They commute with the components L_k in the laboratory system

$$[I_i, L_k] = 0 \qquad \text{for } i = 1, 2, 3, \quad k = x, y, z. \tag{A.15}$$

This means that the components $I_i = \mathbf{J}\cdot\mathbf{e}_i$ (where $\mathbf{e}_i$ are unit vectors along the body-fixed axis) are scalars with respect to rotations in the laboratory frame.

(ii) They obey the commutation relations

$$[I_i, I_k] = -i \cdot I_l \qquad \text{with } (i,k,l) \text{ cyclic } (1,2,3). \tag{A.16}$$

They differ from the normal commutation relations for angular momenta by a minus sign.

The commutators of the operators L_k and I_i with the rotation operator $R(\Omega)$ (A.4) are given by

$$\begin{aligned} [L_x, R] &= -J_x \cdot R, & [I_1, R] &= -R \cdot J_x; \\ [L_y, R] &= -J_y \cdot R & [I_2, R] &= -R \cdot J_y; \\ [L_z, R] &= -J_z \cdot R & [I_3, R] &= -R \cdot J_z. \end{aligned} \tag{A.17}$$

Now we are able to determine the tensor properties of the Wigner functions D^I_{MK} under rotations around the axis in the laboratory and in the body-fixed frame. For the spherical components (A.12), we get

$$[L_\mu, D^{I*}_{MK}] = \sum_{M'} D^{I*}_{M'K} \langle IM' | J_\mu | IM \rangle \tag{A.18}$$

and

$$[I_\mu, D^{I*}_{MK}] = \sum_{K'} D^{I*}_{MK'} \langle IK | J_\mu | IK' \rangle. \tag{A.19}$$

The functions D^{I*}_{MK} therefore behave like spherical tensors of rank I with the magnetic quantum number M under rotations of the laboratory frame. In particular, we get

$$\begin{aligned} &[L_z, D^{I*}_{MK}] = M D^{I*}_{MK} \\ &[L_x \pm iL_y, D^{I*}_{MK}] = \sqrt{I(I+1) - M(M \pm 1)}\; D^{I*}_{M \pm 1 K} \\ &[I_3, D^{I*}_{MK}] = K D^{I*}_{MK} \\ &[I_1 \pm iI_2, D^{I*}_{MK}] = \sqrt{I(I+1) - K \cdot (K \mp 1)}\; D^{I*}_{MK \mp 1}. \end{aligned} \tag{A.20}$$

The normalized wave functions

$$|IMK\rangle = \sqrt{\frac{2I+1}{8\pi^2}}\; D^{I*}_{MK}(\Omega) \tag{A.21}$$

are therefore eigenfunctions of $\mathbf{J}^2$, L_z ($\equiv J_z$), and I_3,

$$\begin{aligned} \mathbf{J}^2 |IMK\rangle &= I \cdot (I+1) |IMK\rangle, \\ L_z |IMK\rangle &= M |IMK\rangle, \\ I_3 |IMK\rangle &= K |IMK\rangle, \end{aligned} \tag{A.22}$$

and form a complete and orthogonal set in the space of the functions of the Euler angles [Ed 57, Chap. IV]:

$$\sum_{IMK} |IMK\rangle\langle IMK| = \delta(\Omega - \Omega'); \qquad \langle IMK|I'M'K'\rangle = \delta_{II'} \cdot \delta_{MM'} \cdot \delta_{KK'}. \tag{A.23}$$

From Eq. (A.17), for a rotation of 180° around the 1-axis we obtain

$$e^{i\pi I_1} D^{I*}_{MK}(\Omega) = \langle IM|R(\Omega)e^{-i\pi J_x}|IK\rangle^* = (-)^I \langle IM|R(\Omega)|I-K\rangle^*$$

or

$$e^{i\pi I_1}|IMK\rangle = (-)^I|IM-K\rangle, \tag{A.24}$$

and similarly,

$$e^{i\pi I_2}|IMK\rangle = (-)^{I-K}|IM-K\rangle. \tag{A.25}$$